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T11 – 05

Assignment No. 5

Aim: Implementation of Statistical Hypothesis Correlation Test

1. What is correlation and covariance?

Both **correlation** and **covariance** measure the relationship between two variables, but they have different properties and interpretations.

1. Covariance

- **Definition:** Covariance measures the direction of the relationship between two variables. It tells us whether the variables move **together** (positive covariance) or in **opposite directions** (negative covariance).

- **Formula:**

$$\text{Cov}(X, Y) = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$$

where:

- X_i, Y_i are individual data points,
- $\bar{X}, \bar{Y}$ are the means of X and Y ,
- n is the number of data points.

- **Interpretation:**
 - **Positive covariance:** Both variables increase or decrease together.
 - **Negative covariance:** When one variable increases, the other decreases.
 - **Zero covariance:** No linear relationship.

- **Limitation:** Covariance is **scale-dependent**, meaning it depends on the units of the variables.
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2. Correlation

- **Definition:** Correlation standardizes covariance, making it unit-free. It measures both the **strength** and **direction** of a linear relationship between two variables.
- **Formula:**

- **Formula:**

$$\rho_{X,Y} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

where:

- σ_X, σ_Y are the standard deviations of X and Y .

- **Interpretation:**
 - **Ranges from -1 to +1:**
 - +1 → Perfect positive correlation.
 - 0 → No correlation.
 - -1 → Perfect negative correlation.
- **Advantages:** Unlike covariance, correlation is **unit-independent**, so it allows for comparisons across different datasets.

2. What is the role of correlation in data science?

Role of Correlation in Data Science

Correlation plays a crucial role in data science, as it helps in understanding relationships between variables, improving model performance, and making data-driven decisions. Below are some key roles of correlation in data science:

1. Feature Selection & Dimensionality Reduction

- Correlation helps identify **highly related features**, allowing for the removal of redundant variables.
- Example: If two features are highly correlated, one can be dropped to avoid **multicollinearity**, improving model efficiency.

2. Identifying Relationships in Data

- Correlation helps determine whether two variables **move together** (positive correlation) or **in opposite directions** (negative correlation).
- Example: In sales analysis, a strong positive correlation between advertising spends and revenue suggests a relationship between the two.

3. Improving Machine Learning Models

- Correlation helps in selecting **important predictors** and removing irrelevant ones, leading to better model accuracy.
- In regression models, highly correlated independent variables can cause **multicollinearity**, which affects model interpretability.

4. Data Preprocessing & Cleaning

- Correlation helps detect **errors or anomalies** in data.
- Example: If age and years of experience have a negative correlation, it might indicate a data entry issue.

5. Time Series & Forecasting

- Correlation is used to measure relationships between past and future values in time series forecasting.
- Example: Stock prices of different companies may be correlated, influencing investment decisions.

6. Market Basket Analysis & Recommendation Systems

- In e-commerce and marketing, correlation is used to find **associations between products**.
- Example: If customers who buy laptops also frequently buy laptop bags, this correlation can be used for cross-selling strategies.

7. Risk Management & Finance

- In finance, correlation helps in **portfolio diversification** by identifying assets that move independently.
- Example: Stocks with low or negative correlation reduce overall risk in a portfolio.

8. Medical & Scientific Research

- Correlation helps analyse relationships between variables in experiments.
- Example: Finding correlations between smoking and lung disease in medical studies.

1. Calculate covariance (Age, Strength) and correlation (Age, strength) for following data using

a. Pearson's Correlation coefficient method

b. Spearman's rank correlation method (use `scipy.stats.pearsonr` and `scipy.stats.spearmanr` function)

Age	38	62	22	38	45	69	75	38	80	32	51	56	21	34	76
Strength	20	15	30	21	18	12	14	28	09	22	20	19	28	23	14

Program:

```
import numpy as np
from scipy.stats import pearsonr, spearmanr
age=np.array([38,62,22, 38,
45, 69, 75, 38, 80, 32, 51, 56, 21, 34, 76])
strength = np.array([20, 15, 30,
```

```

21, 18, 12, 14, 28, 9, 22, 20, 19, 28, 23, 14])
covariance=np.cov(age,strength)[0,1]
print(f"Covariance (Age, Strength): {covariance:.2f}") pearson_corr,
_ = pearsonr(age, strength)
print(f"Pearson's Correlation Coefficient: {pearson_corr:.4f}")
spearman_corr, _ = spearmanr(age, strength)
print(f"Spearman's Rank Correlation Coefficient: {spearman_corr:.4f}")

```

Output:

```

Covariance (Age, Strength): -111.00
Pearson's Correlation Coefficient: -0.9229
Spearman's Rank Correlation Coefficient: -0.9362

```

Conclusion:

The negative covariance (-111.00) and strong negative Pearson (-0.9229) and Spearman (-0.9362) correlation coefficients indicate a **strong inverse relationship** between age and strength. As age increases, strength tends to decrease significantly. The high Spearman value suggests this trend holds even in ranked data.

2. Load the california_housing dataset from scikitlearn. Find the correlation between each of the features using pearson's method and spearman's method. Draw the heatmap for both cases. Identify if any redundant features are there and which independent features are highly correlated with dependent feature. Use following lines of code to import the dataset from sklearn and then create the dataframe from it: from sklearn.datasets import fetch_california_housing housing = fetch_california_housing() **Program:**

```

import pandas as pd import
numpy as np import seaborn
as sns import
matplotlib.pyplot as plt
from sklearn.datasets import fetch_california_housing

```

```

housing = fetch_california_housing()

df = pd.DataFrame(housing.data, columns=housing.feature_names)

df['MedHouseVal'] = housing.target

pearson_corr = df.corr(method='pearson')

spearman_corr = df.corr(method='spearman')

plt.figure(figsize=(14, 7))

plt.subplot(1, 2, 1)
sns.heatmap(pearson_corr, annot=True, fmt='.2f', cmap='coolwarm', vmin=-1,
vmax=1)
plt.title('Pearson Correlation Heatmap')

plt.subplot(1, 2, 2)
sns.heatmap(spearman_corr, annot=True, fmt='.2f', cmap='coolwarm', vmin=-1,
vmax=1)
plt.title('Spearman Correlation Heatmap')

plt.tight_layout() plt.show()
threshold = 0.9
redundant_features = []
for column in
pearson_corr.columns:
high_corr_features =
pearson_corr.index[pearso
n_corr[column] >
threshold].tolist()
high_corr_features.remove
(column) # Remove the

```

feature itself from the list

if high_corr_features:

redundant_features.append((column, high_corr_features))

print("Redundant features (highly correlated with each other):") **for**

feature, correlated_features **in** redundant_features:

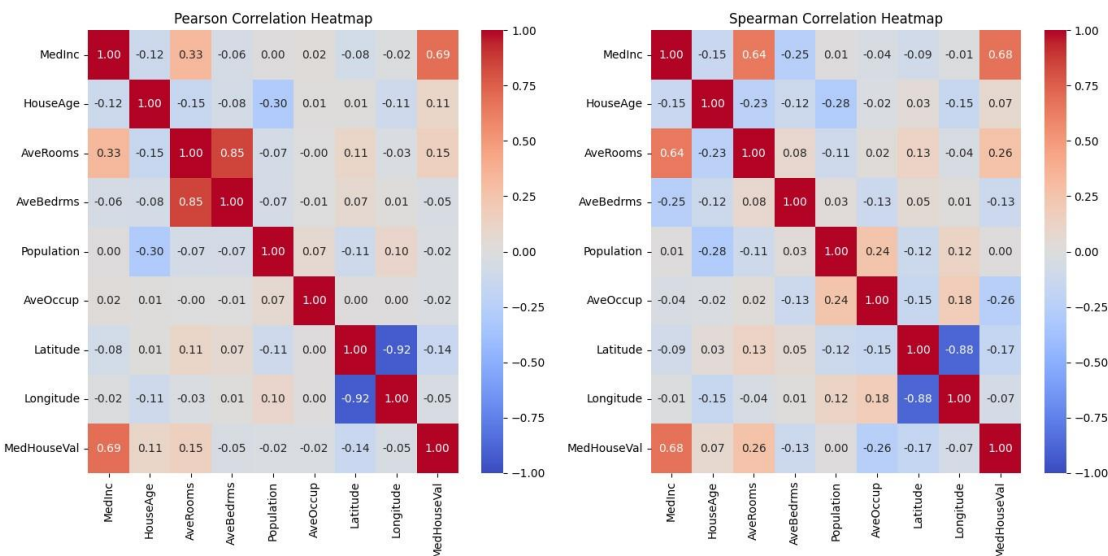
print(f"{feature} is highly correlated with {' '.join(correlated_features)}")

correlation_with_target = pearson_corr['MedHouseVal'].drop('MedHouseVal')

high_corr_with_target = correlation_with_target[abs(correlation_with_target) > 0.5]

print("Independent features highly correlated with the dependent feature (MedHouseVal):")

print(high_corr_with_target) **Output:**



Redundant features (highly correlated with each other):

Independent features highly correlated with the dependent feature (MedHouseVal):

MedInc 0.688075

Name: MedHouseVal, dtype: float64

Conclusion:

The Pearson and Spearman correlation heatmaps reveal key relationships between housing dataset features. **Median Income (MedInc) and Median House Value (MedHouseVal) show a strong positive correlation (~0.69 Pearson, ~0.68 Spearman), indicating that higher income areas tend to have higher house values.** Latitude and Longitude exhibit a strong negative correlation (~-0.92 Pearson, ~-0.88 Spearman), suggesting a geographical trend in housing values. Spearman's correlation captures more nonlinear relationships, as seen in the stronger correlation between **AveRooms and MedHouseVal (0.33 Pearson vs. 0.64 Spearman), highlighting a possible ranking effect**